

Professor Quant Partnership Model

A simulation of an LLM-enabled quant hedge fund as an alternative funding source for university research labs.

Repository: https://github.com/huazuhao/professor_quant_partnership_model

This repository studies a new research-lab funding model made possible by the rapid improvement of LLM coding.

University labs in mathematics, statistics, optimization, simulation, computational modeling, and related STEM fields already have the talent systematic hedge funds rely on. The old bottleneck was implementation: turning a research idea into clean data, backtests, diagnostics, risk controls, deployment code, and monitoring usually took more engineering time than a short sabbatical or research leave could support.

LLM-assisted coding changes that. A 6- to 12-month sabbatical can now be enough for professors to help invent or improve systematic trading strategies. If validated strategies are traded, part of the profits can fund the contributing labs over the following years.

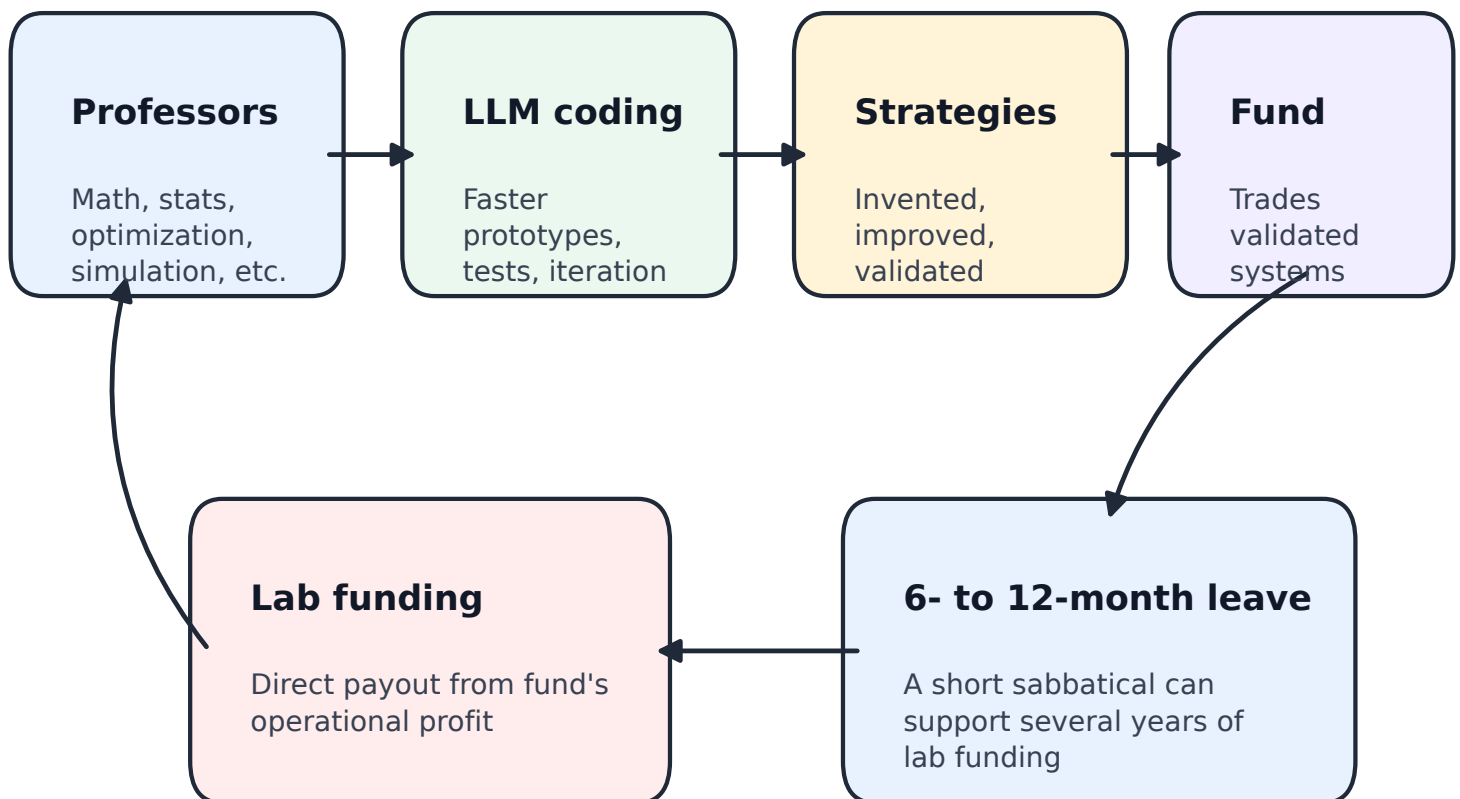
In the current simulation, this loop can start with \$20M of initial capital and grow into a \$1B+ fund. The median 10-year investor outcome is about 3x capital, 120 professor/lab groups receive funding, and five-year lab cumulative compensation is above \$1M at the median.

What the simulation tests

Whether focused professor research, LLM-accelerated implementation, fund infrastructure, strategy profits, and recurring lab support can become a large alternative funding channel for universities.

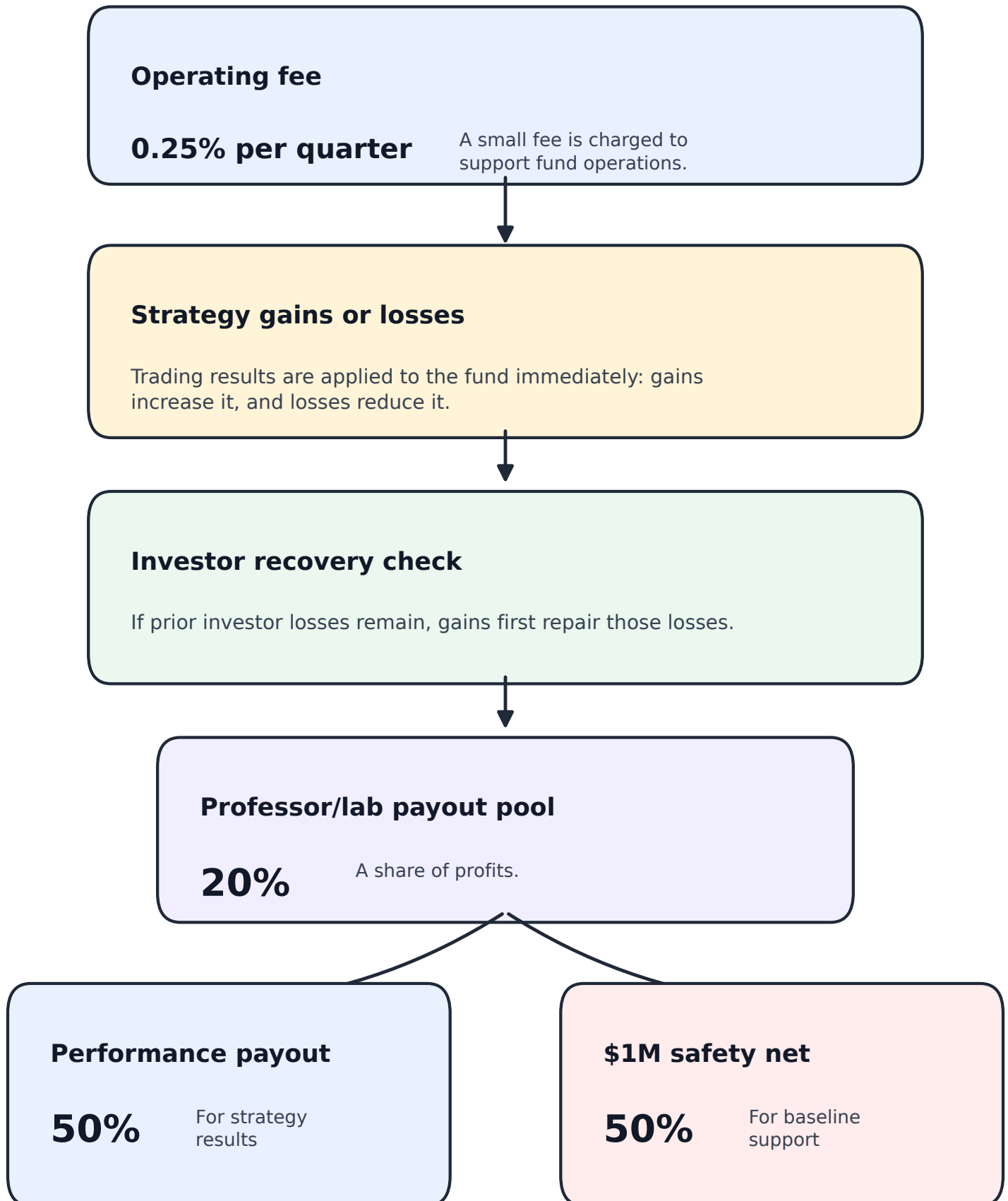
The Funding Loop

A short research leave becomes strategy development, fund infrastructure, and recurring lab support.



Realistic Hedge Fund Accounting

The simulation models the accounting in sequence: operating fee, strategy gains or losses, investor loss recovery, then professor/lab payouts from eligible profits.



Professor/lab payouts are split into two parts: half rewards direct strategy performance, and half funds the safety net for eligible contributors below the \$1M lifetime threshold.

Current Simulation Results

Baseline batch run: 100 paths, 40 quarters, seed 42.

Metric	Current baseline result
Initial fund capital	\$20 million
Final fund size after 10 years	\$1.27 billion median
Median 10-year investor value	\$100 grows to \$302.52
Professor/lab groups receiving funding after 10 years	120 median
Five-year cumulative professor/lab compensation	\$1.34 million median
Five-year cumulative compensation, p75	\$1.63 million
Five-year cumulative compensation, p90	\$2.00 million

Interpretation

These are simulation outputs, not forecasts. The purpose is to test whether the economic loop is internally coherent and whether the scale of the research-funding impact is large enough to justify deeper investigation.

Chart 1: Investor Outcome Distribution

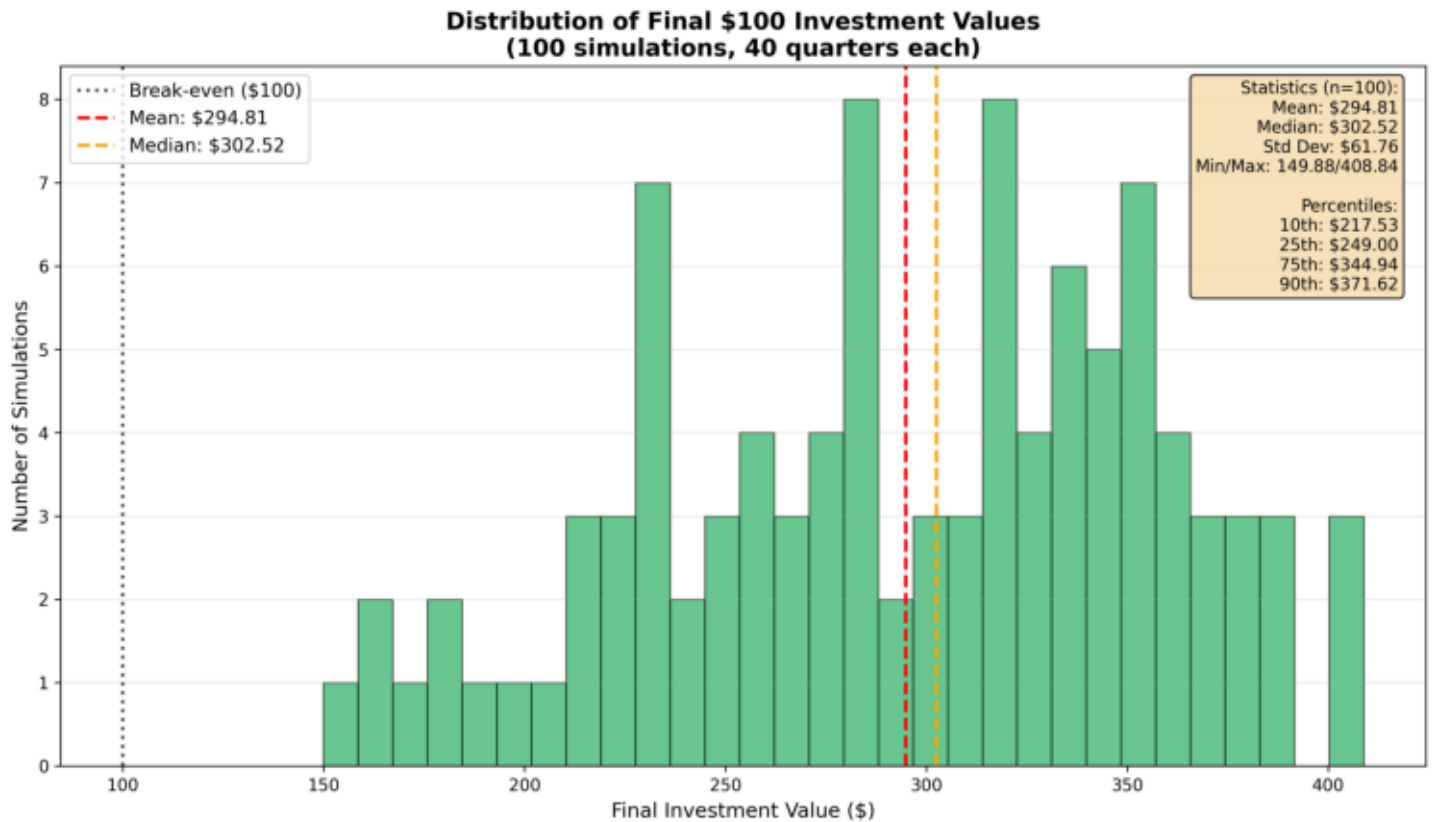


Chart 2: Five-Year Professor/Lab Compensation

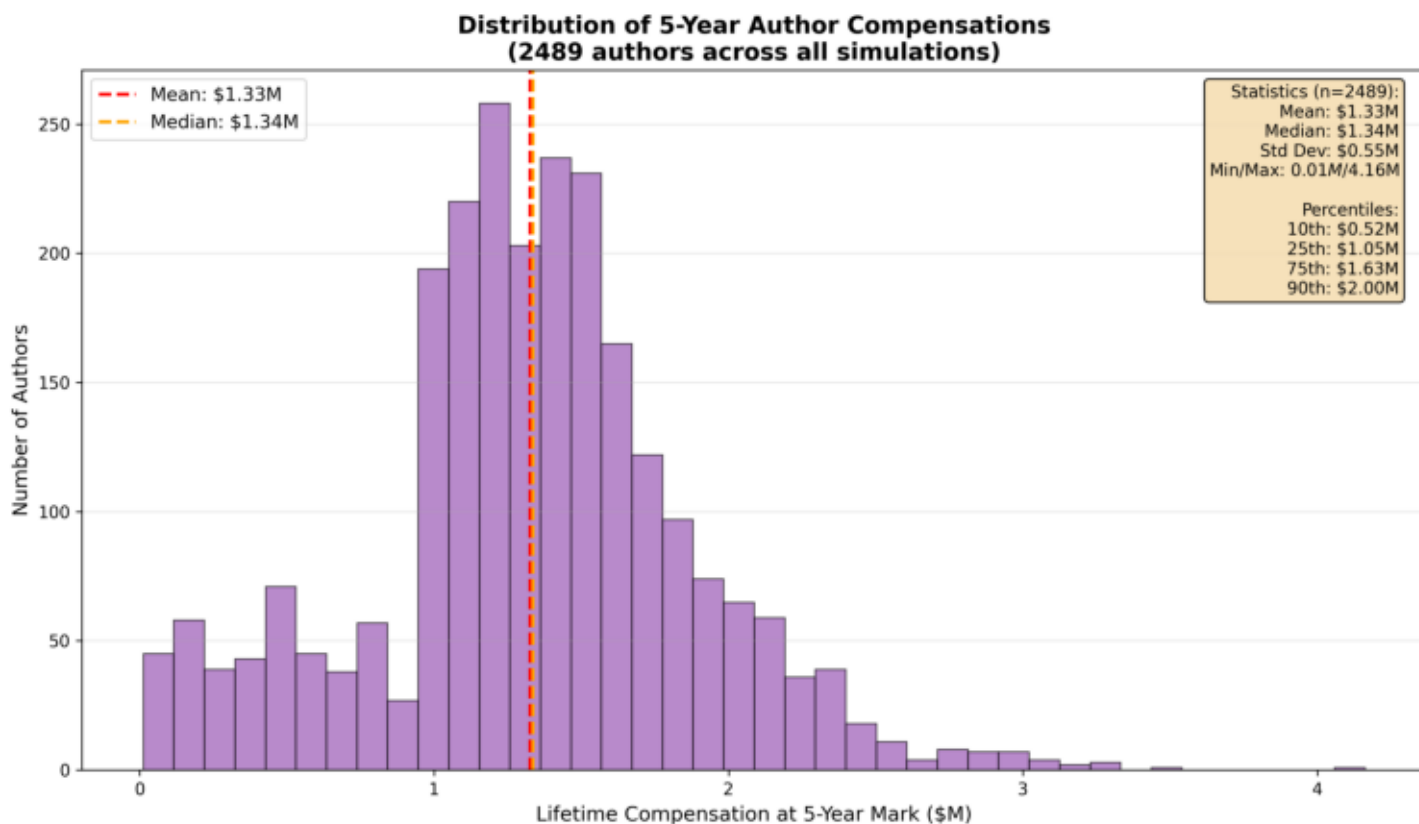


Chart 3: Final Fund AUM

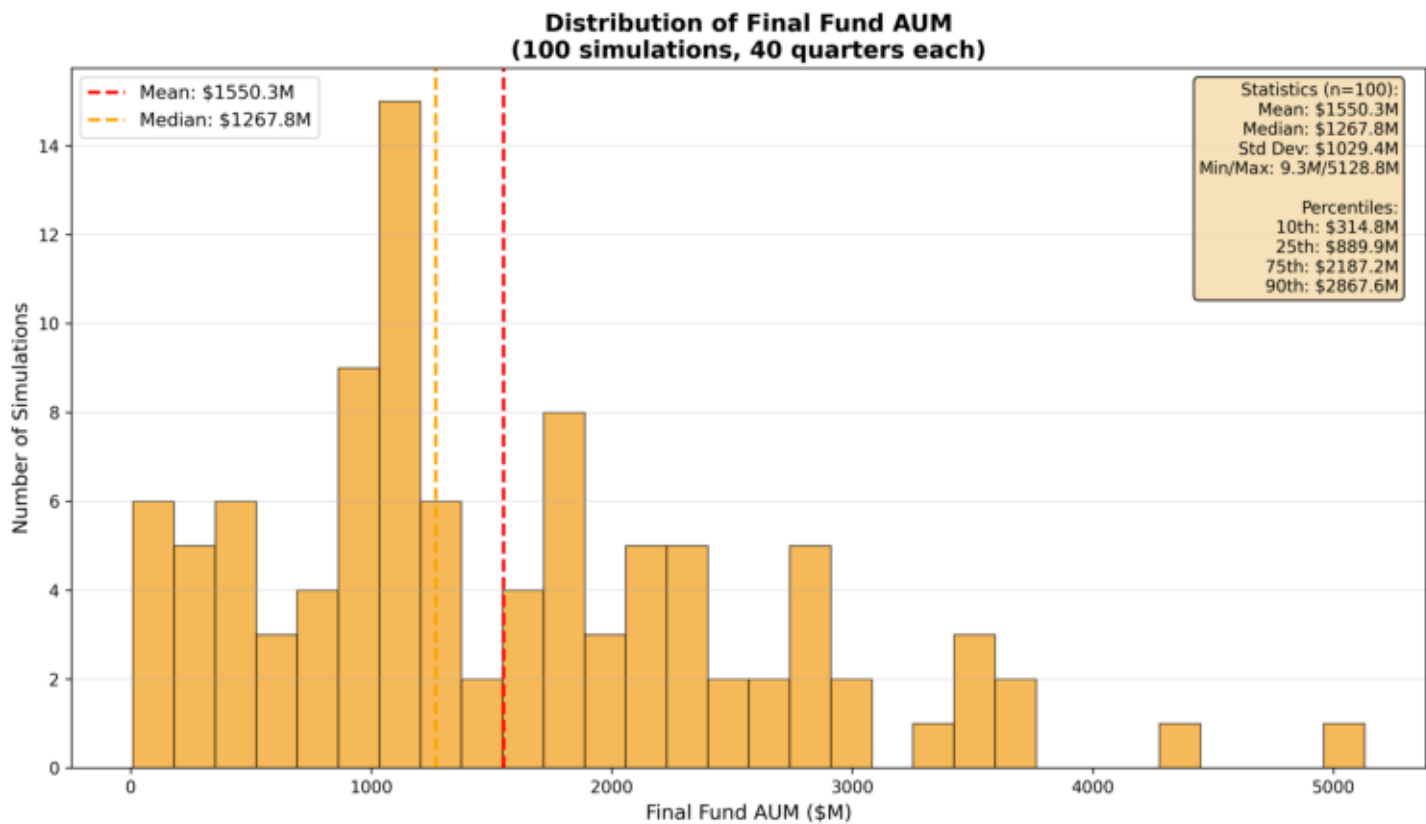


Chart 4: Professor/Lab Groups Receiving Funding

